

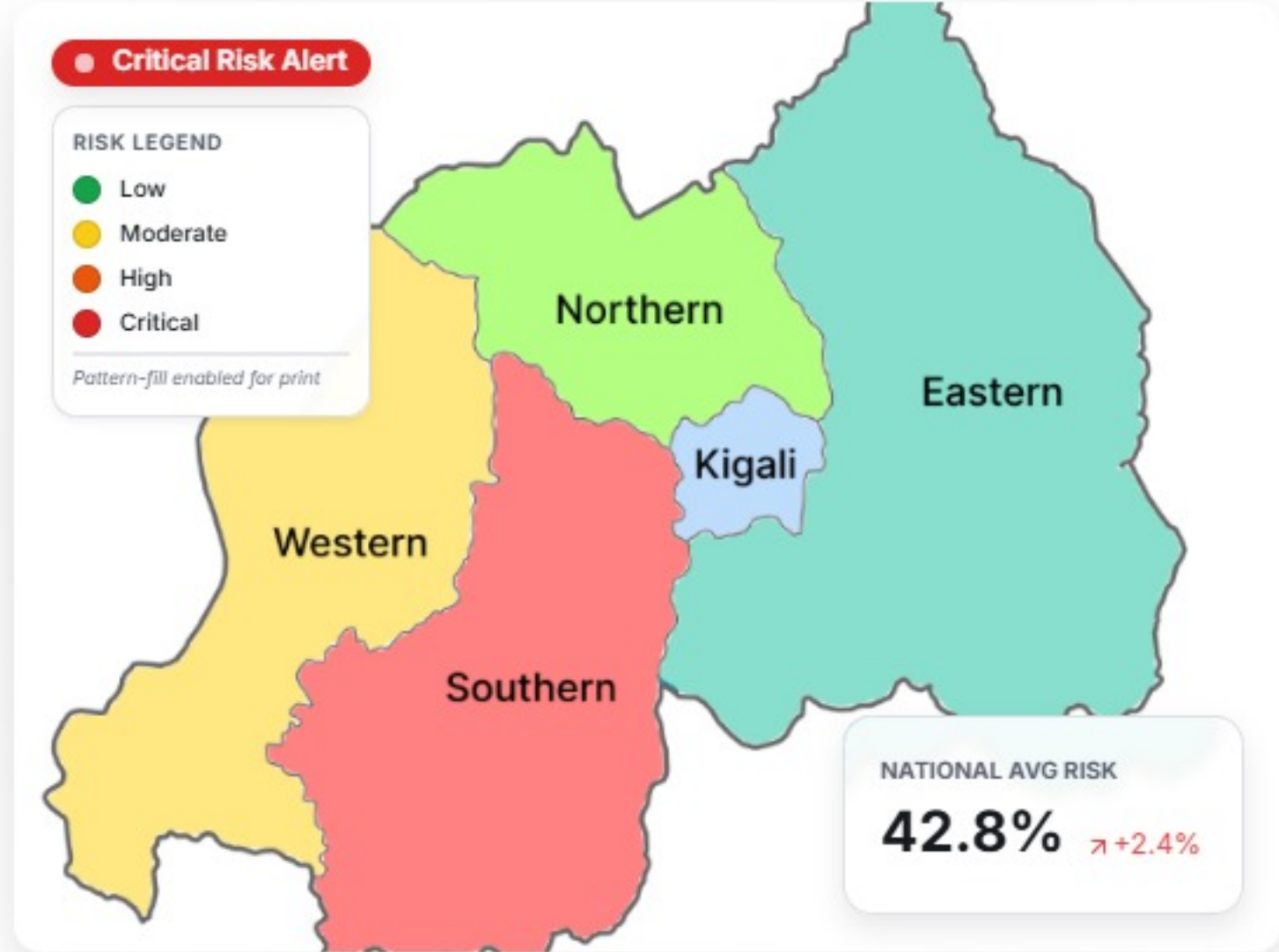
Powered by AI & Satellite Data

Climate Intelligence for a Malaria-Free Rwanda.

Predictive risk mapping for climate-driven health crises. Zero Bite gives you the data to act before the outbreak.

Q Enter your district (e.g., Kayanza)

Check Risk



Trusted by 30+ District Health Officers across Rwanda.

How Zero Bite Works

The end-to-end intelligence pipeline protecting Rwandan communities from climate-related health disasters.



Data Ingestion

We pull real-time satellite imagery and Meteo Rwanda weather feeds.



AI Prediction

Our models process 12+ indicators to predict malaria breeding risk 30 days ahead.



Actionable Response

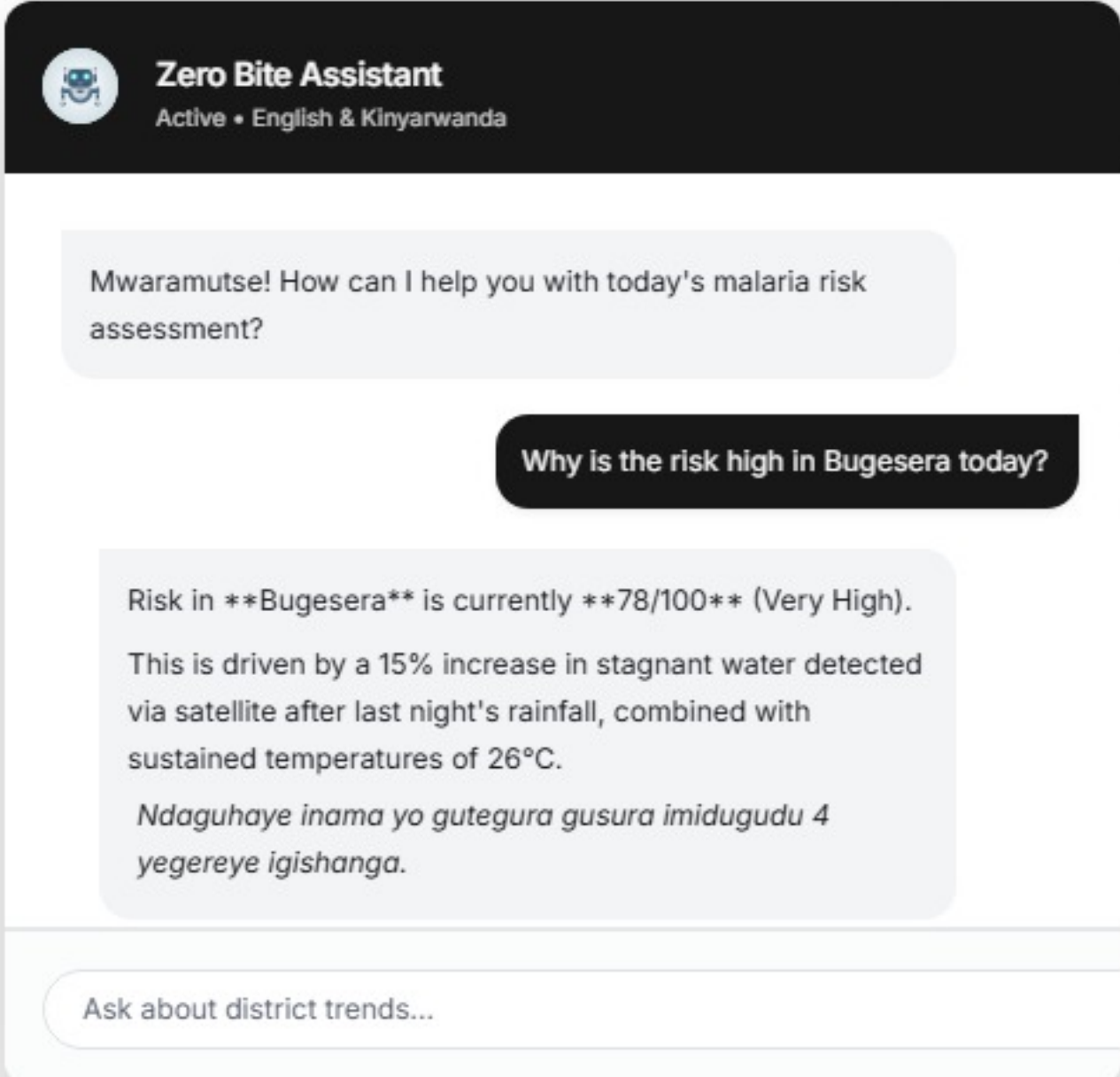
Localized alerts are sent to CHWs and Ministry officials for immediate intervention.

Virtual Assistant

Your 24/7 Climate Intelligence Partner

The Zero Bite AI Assistant understands the complex relationships between rainfall, temperature, and vector breeding. Get instant insights in your language.

- ✓ Ask about specific district forecasts
- ✓ Get recommendations for community health visits
- ✓ Draft SMS alerts in Kinyarwanda or English
- ✓ Understand the data behind risk scores



Who Zero Bite Is For

Tailored dashboards for every level of the health system.



Ministry Officials

MINISTRY

National level strategic overview and cross-district resource allocation.

Enter Dashboard →



District Health Officers

DISTRICT

Deep-dive district analytics, sector mapping, and 14-day forecasts.

Enter Dashboard →



CHWs & Village Leaders

CHW

Immediate action plans, SMS alert workflows, and local observation logs.

Enter Dashboard →



General Public

PUBLIC

Personal risk scores, prevention tips, and localized weather snapshots.

Enter Dashboard →

Prevention at a Glance

Immediate steps to take based on current national climate trends.

Mosquito Nets

Ensure all household members sleep under insecticide-treated nets.

Clear Water

Empty or cover all containers of standing water near homes.

Bush Clearing

Keep grass short and clear dense vegetation around dwellings.

Peak Exposure

Avoid being outdoors during dusk and dawn peak biting hours.

View full health guide →

Subscribe to Local Alerts

Receive real-time SMS alerts in English or Kinyarwanda when risk levels in your district increase.

Full Name

+250 XXX XXX XXX

Select District

Subscribe Now



Zero Bite is a free service provided in partnership with the Ministry of Health. Standard SMS rates do not apply. You can unsubscribe at any time.

Trust Through Science

Our models are trained on 15 years of historical Rwandan epidemiological data and validated against high-resolution satellite imagery from the European Space Agency.

94%

PREDICTION ACCURACY

12m

DATA LATENCY

30+

DISTRICTS MONITORED

OUR RESEARCH & DELIVERY PARTNERS



2023 National Climate-Health Report

Download the comprehensive analysis on how AI is transforming outbreak prevention in Rwanda.

Download PDF (14.2 MB)

Ready to secure your community?

Contact our team for specialized training, API access, or to join the District Health Intelligence Network.

Contact Our Team

View Public Portal



Zero Bite

Protecting communities through satellite-driven AI predictions and real-time climate intelligence.

Resources

Data Methodology
Meteo Rwanda Portal
MoH Health Guide

Support

Help Center
Contact Us
Admin Login

Platform

Public Dashboard
SMS Subscription
API Access



Welcome to Zero Bite

Please select your designated role to access the climate-intelligence portal.

**Community Worker**

Village-level monitoring & logs

**District Officer**

Regional forecasts & alerts

**Ministry Official**

National strategy & risk mapping

**Public Portal**

General risk view (Read-only)



Authorized access only. By signing in, you agree to the Rwanda Data Protection guidelines and Ministry of Health security protocols.

Community Worker Authentication

Secure Login

Enter your credentials below to access the platform.



Phone OTP



Email & Pass

Phone Number (MTN/Airtel)



+250 788 000 000

A one-time verification code will be sent to this number.

Send OTP Code

SYSTEM RELIABILITY

**Offline Sync & SMS Fallback** **ACTIVE**

Low connectivity? Zero Bite supports SMS-based data logging and critical alert reception. Your dashboard will sync automatically when back online.

[← Back to Information Portal](#)

- Dashboard
- Alerts Center
- District List
- Reports
- Settings

National Climate Intelligence

Strategic oversight and AI-driven predictive modeling for Rwandan Districts.

Export Report

All Hazards



High-Risk Districts
8 / 30 districts



Active Warnings
12 Disaster alerts active



Avg. National Risk
42 +4% from last week



Population at Risk
1.2M Estimated reach

District Risk Heatmap

Risk Score

Alert Density

LEGEND

- Critical Risk
- High Risk
- Moderate
- Low Risk

Toggle Hazard Layers

Sync Satellite Data

CHOROPLETH VIEW ACTIVE • 30 DISTRICTS TRACKED



TIMELINE FORECAST

Oct 24 — Nov 07

LIVE

AI Situation Summary

v2.4 Engine

Generated based on current satellite feed

Summary: Increased humidity in the **Eastern Province** combined with stagnant water data suggests a **high probability (78%)** of malaria spike in the Kayonza sector within 14 days.

Current national interventions are **on track** for 22 districts, but immediate awareness campaigns are recommended for Bugesera due to rising heat index.

RECOMMENDED ACTIONS

- Deploy SMS Warning kayonza-01**
Targeting 45,000 residents via CHW network.

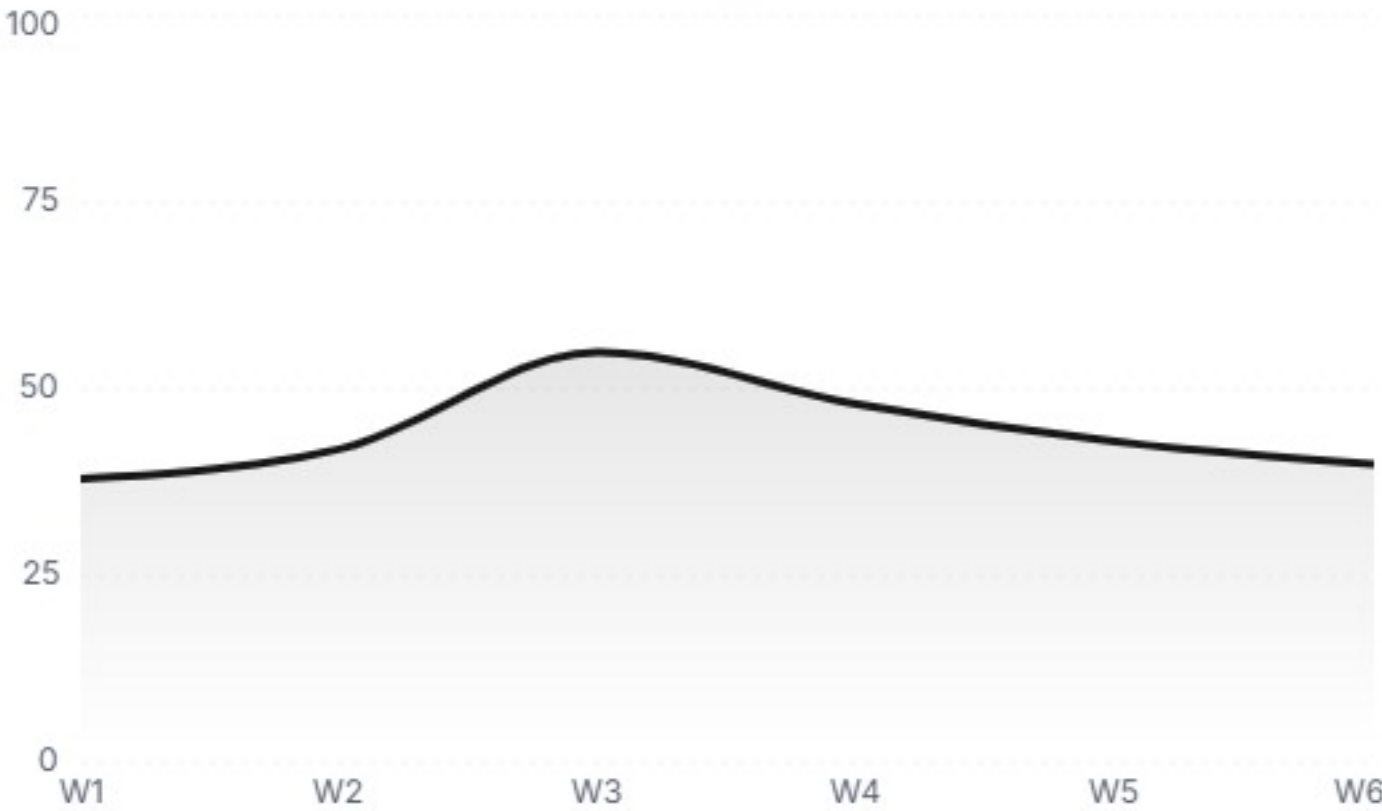
Escalate Bugesera Status
Heatwave threshold reached for 48h period.

Generate Full Strategy Report

Risk Probability Trends

Comparing historical norms vs. predicted intelligence

HISTORICAL PREDICTED



District Priority Ranking

View All Districts >

Top districts requiring immediate attention

DISTRICT	RISK SCORE	HAZARD TYPE	
Kayonza	88	Malaria	>
Bugesera	82	Heatwave	>
Gicumbi	75	Floods	>
Nyamasheke	68	Malaria	>
Rubavu	64	Floods	>

AI Status

Models updated 12m ago. Satellite feed active.

Logout

Dashboard

Alerts Center

District List

Reports

Settings

Kayonza Eastern Province

Find Sector Configuration

High Risk

Last updated: Oct 28, 14:30

Kayonza District

Malaria breeding risk has increased by 14% this week due to unseasonal rainfall in the Eastern Province.

78 / 100

RISK SCORE

Temp: 28°C (+2°)
Humidity: 82% (+12%)
Active Hotspots: 4

Dispatch Sector Alerts

Generate Monthly Report

Sector Risk Heatmap

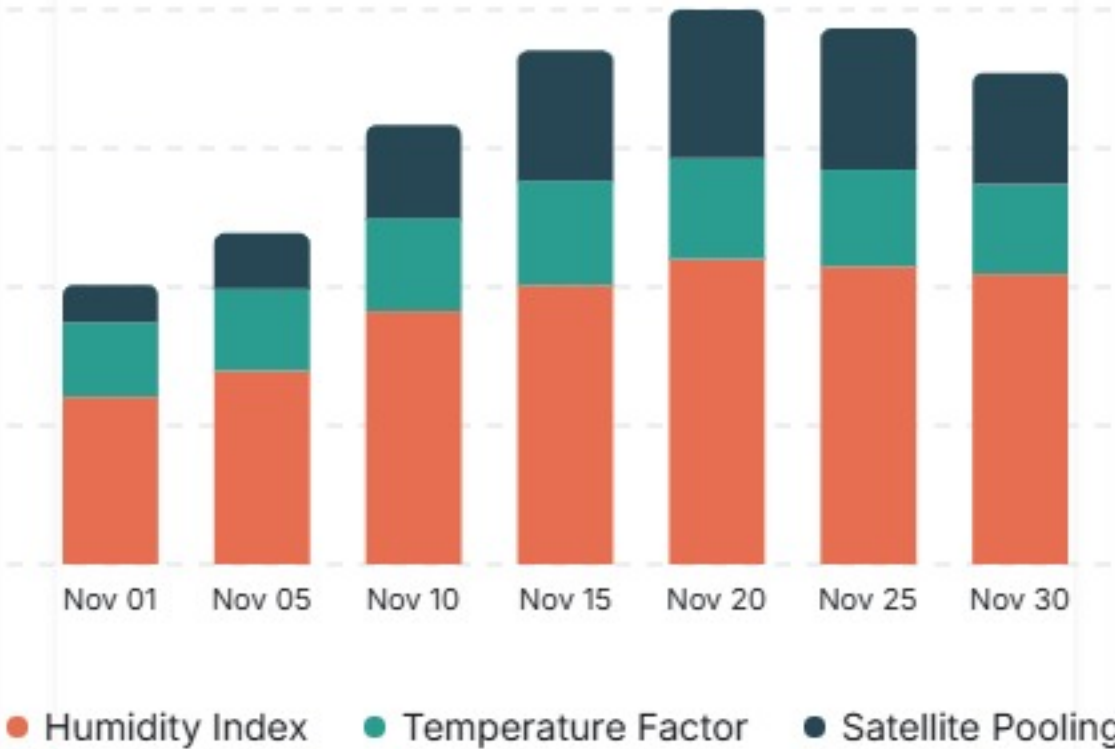
Visual distribution of breeding risk across Kayonza



30-Day Risk Forecast

Contributing factors for predicted malaria breeding

AI MODEL V4.2



Climate Intelligence Advisor

Hello Officer Ndayisaba. I've analyzed the latest satellite data for Kayonza.

The risk score in Mwiri and Ndego sectors has jumped to 91. This is primarily driven by persistent humidity levels above 80% and new surface water pooling detected near the Akagera river banks.

AI Assistant • 2:30 PM

RECOMMENDED ACTIONS

- Deploy larvicide teams to Ndego Sector marshes.
- SMS Alert: Advise residents near Mwiri on ITN usage.
- Audit ITN stock at Gahini District Hospital.

SATELLITE ANALYSIS

Recent Sentinel-2 imagery indicates vegetation growth in abandoned quarries, creating micro-habitats for Anopheles larvae.

Recent Alerts Dispatch

- SMS Alert Kayonza CHWs 2h ago
Heavy pooling detected. Initiate larval source
- Disaster Warning Ndego Sector 5h ago
Flood risk level orange. Evacuate low-lying areas.
- System Update All Staff 1d ago
Satellite feed updated. Risk scores recalculated.

View All Dispatch Logs

Resource Availability

ITN STOCK 12.4k

SPRAY TEAMS 08

Community Outreach Goal 68%

Request Logistics Support

AI Status

Models updated 12m ago. Satellite feed active.

Logout



Worker Dashboard

Kinyinya Village, Gasabo District Online

Recent Logs

Complete Session

VILLAGE RISK LEVEL

84

High Risk Identified

Based on satellite moisture & rainfall data from the last 24h.

Today's Goals

- Contact 12 Village Leaders
3/12 completed
- Distribute SMS Alert
Broadcasted at 08:45 AM

Weather Warning

Heavy rains expected in Gasabo District within the next 48 hours. Ensure all stagnant water logs are closed before Tuesday.

Immediate Community Warning

Priority 1

High-risk breeding detected in North Sectors. Public broadcast required.



Satellite Alert: Anomaly Detected

Humidity levels reached 82% threshold for vector proliferation.

Compose Broadcast

View Sector Map

Broadcast Community SMS

Send warnings or health advice to all registered phones in your cell.

1,240 Contacts

Warning

Prevention

Visit

BITEYE: Inzu yawe iri mu gace gafite ibyago byinshi by'imibu. Siba amazi adatemba kandi ukoreshe inzitiramibu. - Zero Bite

122 / 160 characters



AI Suggestion: This message is highly effective. Translation in Kinyarwanda is accurate for the local context.

Schedule

Send Community SMS

Field Monitoring Tools



Log Observation



Check Water Site



Schedule Visit



Share Report

Why is the risk High today?

Satellite sensors from Meteo Rwanda have detected abnormal rainfall patterns followed by a warm dry spell. This combination creates optimal breeding conditions for Anopheles mosquitoes in the marshland areas of Kinyinya.

SMS Guidelines for Village Leaders

Alerts Inbox

3 New

Urgent

10m ago

Visit Cell B Breeding Site

Satellite data shows 45% increase in moisture. Check for stagnant water.

Details

Warning

2h ago

Medication Refill Alert

Malaria prophylactic stock at Village Center is below 20%.

Details

Info

5h ago

Community Meeting

Discuss net distribution for next Monday with the Cell Leader.

Details

View All Alert History

Community Profile



Cyprien Mutabazi

Cell Leader (Umukuru w'Umudugudu)



Grace Ishimwe

Health Coordinator

Contact Tree



Need Advice?

The AI Assistant can help with situational questions and Kinyarwanda translation.

Ask Assistant

Check Your District's Climate Risk

Real-time satellite insights for malaria prevention across Rwanda's 30 districts.

Kigali - Gasabo

Popular:

Kigali

Musanze

Rubavu

Huye

Kigali - Gasabo

Active Forecast

68

High Risk

Updated: Today, 08:00 AM

AI Prediction Detail

Risk is **High** for the next 7 days. Recent humidity spikes and local vegetation growth have accelerated breeding cycles.

TRANSMISSION STAGE

Acceleration

CONFIDENCE SCORE

94% Accuracy

Local Weather

24°C

Temperature

78%

Humidity

Heavy rainfall recorded in the last 48 hours. Stagnant water levels are increasing.

Risk Band Guide	
Low	0-25 Minimal breeding conditions observed.
Elevated	26-50 Conditions favorable for mosquito activity.
High	51-75 High probability of malaria transmission.
Critical	76-100 Immediate preventive action required.

Recommended Prevention

See All Tips →

Use Bed Nets

Priority

Ensure all family members sleep under insecticide-treated nets.

Clear Water

Environment

Empty containers and clear stagnant water around your dwelling.

Close Windows

Daily

Keep windows and doors closed or screened after 6:00 PM.

Seek Care

Health

Visit your CHW immediately if you develop a sudden fever.

Zero Bite utilizes high-resolution satellite data from Meteo Rwanda and historical health records to generate these models. Risk levels are advisory; always consult local health professionals for medical guidance.

Stay Protected

Get weekly malaria risk alerts for your village via SMS. No smartphone required.

Enter phone number (e.g., 078...)

Subscribe Free

By subscribing, you agree to receive climate health alerts from Zero Bite. Standard rates may apply.

Health Support

Emergency Toll-Free

114

Malaria Hotline

3456

Locate Nearest Clinic



Dashboard

Alerts Center

District List

Reports

Settings

AI Status

Models updated 12m ago. Satellite feed active.

Logout

Alerts Inbox

Search alerts...

All Sent Scheduled Failed

Malaria Risk Spike: Musanze

All CHWs, Musanze District



12 Oct, 10:45 AM

Sent

Seasonal Rainfall Forecast

General Public, Northern Province



12 Oct, 09:00 AM

Scheduled

Flood Alert: Ruhengeri Sector

Emergency Contacts



11 Oct, 04:30 PM

Failed

Monthly Health Briefing

District Health Officers



10 Oct, 02:15 PM

Sent

Hygiene Campaign: Sector 4

Village Leaders



09 Oct, 11:00 AM

Draft

+ Compose New Alert

Alert Composer

Draft climate intelligence messages for cross-channel delivery.

Save as Draft

Send Alert

Compose Message

Templates

Fallback Rules

Delivery Channels & Targeting



Dashboard

Immediate notification in-app for logged-in users.



SMS Broadcast

Direct to mobile. Critical for offline areas.

TARGET AUDIENCE

Select roles or groups



GEOGRAPHIC SCOPE

Select districts/sectors



Message Content

AI Refine

ENGLISH (DEFAULT)

Attention: High rainfall predicted for Musanze over next 3 days. Community Health Workers are advised to increase mosquito net distribution and clear stagnant water sites.

156 / 160 characters

KINYARWANDA (TRANSLATED)

Translation will appear here...

AI Wording Assistant

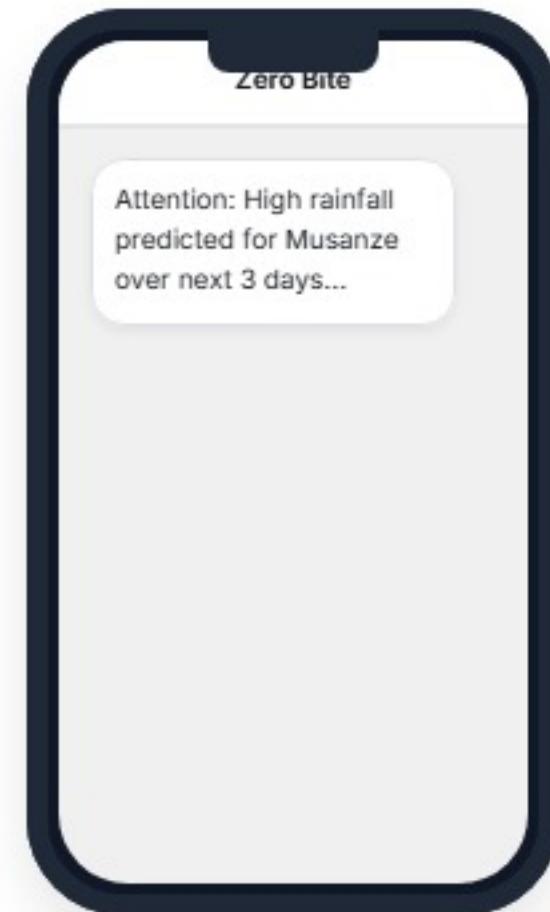
Based on current satellite data in Musanze, I recommend focusing on stagnant water prevention.

"Focus on clearing gutters and stagnant w

"Increase surveillance for fever symptoms

See more suggestions

SMS Preview



- Dashboard
- Alerts Center
- District List
- Reports
- Settings

Ministry > District Management

Administrative Districts

Search districts by name or province...

District Name	Province	Current Risk	7-Day Trend
Gasabo	Kigali	82	Increasing
Kicukiro	Kigali	45	Stable
Nyarugenge	Kigali	31	Decreasing
Musanze	Northern	78	Increasing
Rubavu	Western	64	Increasing
Huye	Southern	22	Decreasing
Nyagatare	Eastern	91	Increasing

AI Status

Models updated 12m ago. Satellite feed active.

Logout

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DISTRICT PROFILE

Gasabo District

AI Risk Prediction

82/100

Increasing

High vegetation indices and sudden rainfall increase predicted for next 14 days. Malaria breeding potential at seasonal high.



Active Data Feeds

Draft Specific District Alert

PDF Report

CSV Export

- Dashboard
- Alerts Center
- District List
- Reports
- Settings

Reports & Analytics

Custom Report Builder

Synthesize district health indicators, climate trends, and AI forecasting into production-ready PDF or CSV exports for stakeholder reporting.

View Analytics Dashboard

- QUICK TEMPLATES
- Monthly Summary
Detailed view of the last 30 days.
- Seasonal Analysis
Climate trends for the current season.
- Annual Brief
High-level executive summary of the

REPORT CONFIGURATION

Time Range

2024-05-012024-05-31

District Filter

KigaliMusanze+ 28 More

Data Indicators

☒ Malaria Risk Index

☒ Disaster Probability

☐ Resource Availability

☒ CHW Intervention Logs

☐ Meteorological Data

Generate Preview

Document Preview

Live visualization of your generated export parameters.

PrintShareExport PDFExport CSV



REPUBLIC OF RWANDA

MINISTRY OF HEALTH • MALARIA DIVISION

Climate Health Intelligence Platform (Zero Bite)

ID: ZB-2024-MAY-04

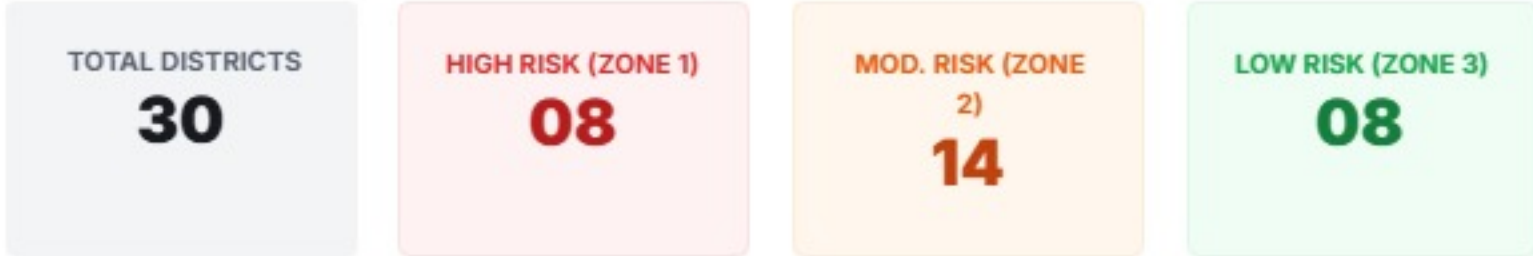
Generated on: May 24, 2024

Reporting Period: May 1 - May 31, 2024

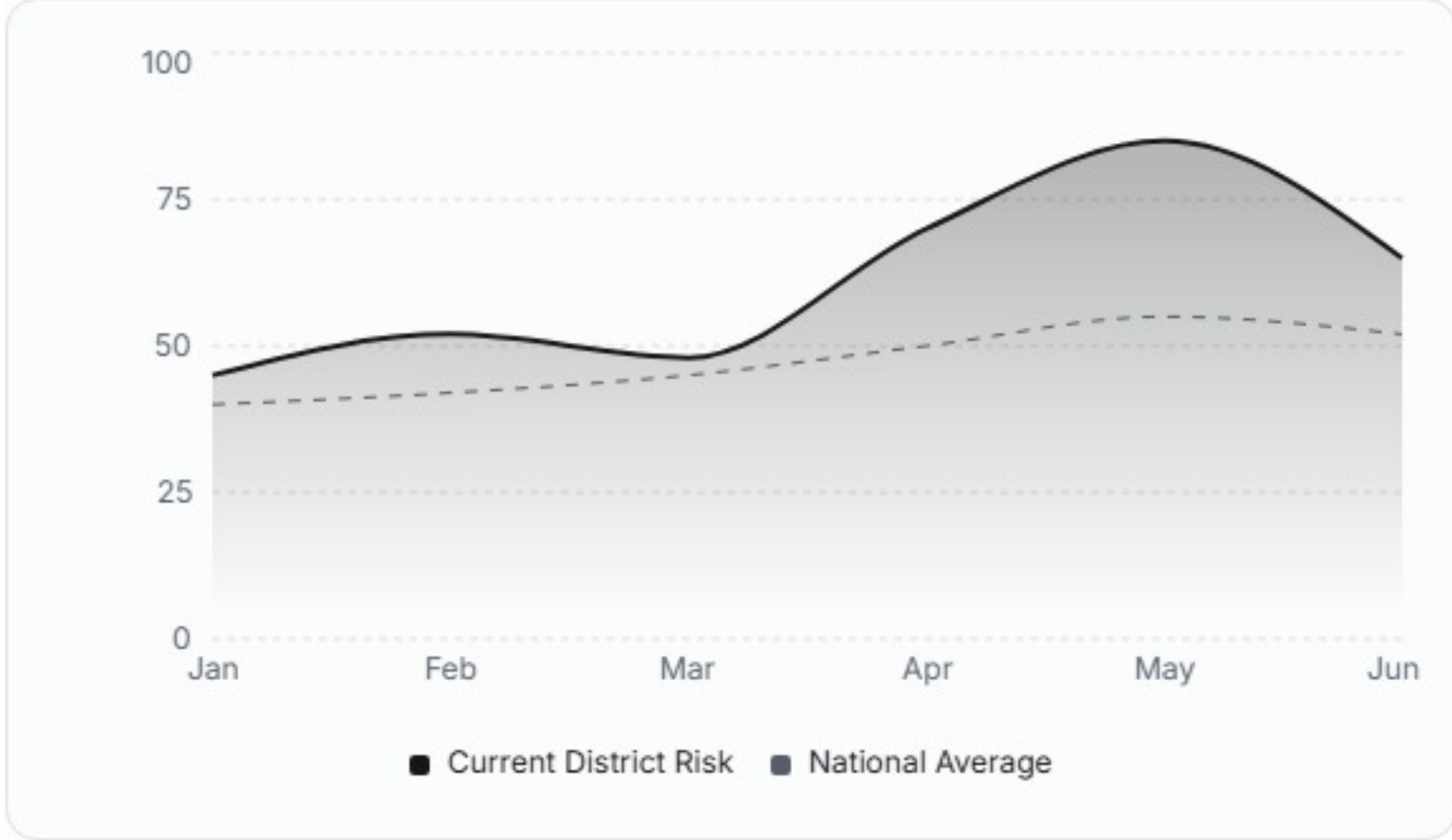
Executive Summary

The month of May 2024 has shown a moderate increase in malaria breeding risk across the Northern and Eastern provinces. AI projections indicate a **12.4% rise** in district-level risk compared to the previous reporting period, largely attributed to higher-than-average precipitation and humidity levels recorded in the sectors of Musanze and Nyagatare.

District Risk Comparison



6-Month Trend Analysis



Indicator Breakdown

DISTRICT	MALARIA RISK	DISASTER PROB.	ACTIVE ALERTS	STATUS
Musanze	85%	12%	3	Urgent
Nyagatare	72%	5%	1	Monitor
Rubavu	44%	28%	2	Stable
Gasabo	31%	2%	0	Stable
Rwamagana	59%	8%	1	Monitor

AI Status

Models updated 12m ago. Satellite feed active.

Logout



- Dashboard
- Alerts Center
- District List
- Reports
- Settings

System Administration / users

Settings & Admin

Manage platform security, technical integrations, and AI risk logic.

- Users
- SMS Gateway
- Thresholds
- Audit Log
- Localization

+ Add User

User	Role	District	Status	Last Active	Actions
Jean-Pierre Kabera	Ministry	National	Active	2 mins ago	Edit
Marie Claire Uwase	District Officer	Musanze	Active	1 hour ago	Edit
Emmanuel Gisa	CHW	Kayonza	Inactive	2 days ago	Edit
Sonia Mukamanzi	Admin	Kigali	Active	Just now	Edit
Aimable Rugamba	District Officer	Nyamagabe	Active	5 hours ago	Edit



System Health & Compliance

Zero Bite complies with Rwanda MoH data privacy standards. All administrative changes are logged for seasonal review.

- Download Security Report
- Contact System Support

AI Status

Models updated 12m ago. Satellite feed active.

Logout